Arjun Mauji

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Education

B.A.Sc. Mechanical Engineering, Queen's University *Kingston, Canada | 2027*

- Dean's Scholar 24/25 & 25/26, Biggs – Ronald & Deanna Scholarship
- Relevant Coursework/Concepts: Embedded Programming, Microcontrollers, Numerical Methods for Mechanical Engineering, Digital Systems for Mechatronics, Material Sciences, Dynamics & Vibrations, Solid Mechanics

Work Experience

Teaching Assistant (Mechatronics Course), Queen's University Sep – Dec 2025

- Co-supervised 10 mechatronics labs with over 100 students on measurement sensors and signal noise filtering
- Answered electronic measurement theory questions and troubleshot lab circuits

Projects

Signal Displayer, Personal Project | GitHub Jul 2026

- Designed a **STM32**-based signal displayer to visualize both PWM and DC signals between 0-3.3V
- Programmed a custom **8 bit SPI driver** to communicate with a TFT display module and draw the signals
- Built an understanding of the **ARM Cortex-M** architecture through manual configuration of hardware peripherals

Embedded Team, Queen's Off-Road Car Design Team (Baja) Feb – Apr 2026

- Programmed **STM32** boards in C to monitor the vehicle transmission system's thermal condition
- Assisted in migrating the vehicle's existing **CAN** bus communication library to use **FreeRTOS**

Hardware Motion Analyzer, Personal Project | GitHub Nov – Dec 2025

- Prototyped a low-cost (<$40) wireless motion analyzer to compare dynamic movement flexibility between users
- Programmed 2 **IMU**-based sensor modules to capture movements by calculating joint flexion and extension angles
- Implemented **RTOS** tasking with **I2C** communication to log and filter up to 10 angle data streams concurrently
- Analyzed movement flexibility and peak angle values using **SciPy**

Two Stage Shifting Gearbox, Team Project | GitHub Jan – Apr 2026

- Built a **2-stage shifting gearbox** and housing to maximize torque and speed for an RC car's climb and sprint modes
- Performed **AGMA fatigue analysis** and iterated gear and shaft geometry to achieve fatigue safety factors above 1.0
- Adapted an initial steel design for 3D printing and modeled the gearbox assembly in **SolidWorks**

Mechanical Team, Queen's Off-Road Car Design Team (Baja) Oct – Dec 2024

- Assisted in the **modelling and manufacturing** of the continuously variable transmission (CVT) using SolidWorks
- Used **FEA** to assess the system's previous cooling methods and identify weight-reduction opportunities

Skills

Mechanical: SolidWorks CAD & FEA, Ansys Fluent & Workbench, XLFR5
Embedded Systems: STM32, ESP32, FreeRTOS, CAN, SPI, I2C
Software Tools: C, Python, MATLAB (Simulink), Git/GitHub, GNU Debugger (gdb)
Electrical Tools: Multimeters, Soldering, Oscilloscopes
Spoken Languages: French (Fluent)